

Explicit Viability Bounds for Safety Filters with Configuration-Dependent Braking Authority

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Abstract—A team of quadrotors carrying a cable-suspended payload may need to reposition before it can brake near an obstacle. If all cables pull toward a wall, the team cannot slow the payload until the cable directions change. This delay can cause a safety filter to accept a control input now and become infeasible later. We derive a condition for this failure in a wall-distance high-order control barrier function (HOCBF) filter. We address the delay using the analytically computable stopping distance of a feasible swing-and-brake maneuver. The maneuver reserves thrust for repositioning while supporting the payload’s weight. Under the stated assumptions, the resulting certificate defines a controlled-invariant set for the continuous-time taut-cable model. The proposed safety filter remains feasible throughout this set. A complementary optimistic model identifies states from which no admissible controller can prevent wall contact. Simulations compare the proposed filter with HOCBF filters. Of 100 trials started at its boundary with a hold of 1 ms, only one reaches the wall, by 2 micrometers, and none does with a hold of 0.5 ms. The certificate is conservative where a swing is required, and a sampled implementation with slack cables lies outside its hypotheses and needs an empirical margin.

I. INTRODUCTION

Cooperative aerial transport uses multiple quadrotors to carry a payload through cables. A cable can pull the payload but cannot push it, so the team’s available braking forces depend on cable directions. If every cable pulls toward a wall, the team must reposition before it can slow the payload (Fig. 1), and safe braking must account for its duration.

Safety filters restrict a nominal controller’s commands to enforce state constraints. A wall-distance high-order control barrier function (HOCBF) filter can admit a command now and lose feasibility later; that is, no command may satisfy both its constraint and the input limits. We ask how much wall clearance is needed when braking requires a prior configuration change.

A. Related Work

High-order and exponential control barrier functions address safety constraints whose inputs enter through higher derivatives [1]–[3]. Input-constrained formulations construct safety conditions that account for limits on the available control [4]. Additional feasibility constraints can preserve admissible control choices over time [5].

Backup CBFs construct controlled-invariant sets using the future motion under a known backup controller [6], [7]. Constructive and predictive barriers also use future trajectories

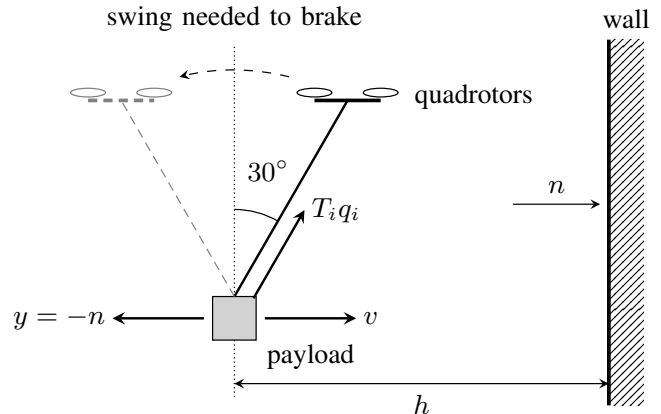


Fig. 1. Two quadrotors carry a payload toward a wall. With both cables leaning toward the wall, tensions cannot brake it; a cable must first swing to the far side (dashed).

to formulate safety conditions [8], [9]. Backup methods have been applied to high-speed drone geofencing [10].

For cable-driven robots, wrench-feasibility analyses characterize the forces available at a given geometry [11], [12]. Cooperative aerial-transport models describe the dynamics that govern changes in cable geometry [13], [14], and CBF-based planning has been developed for obstacle avoidance during cooperative payload manipulation [15].

Shi et al. address the time needed for a quadrotor to reorient its thrust before braking [16]. Their escape-aware barrier accounts for reorientation limits imposed by bounded body rates. The cable-transport setting considered here couples braking forces to cable coordinates with bounded second-order dynamics. Our focus is an explicit calculation of the future wallward excursion caused by this reconfiguration delay. The calculation supports complementary sufficient and necessary conditions for avoiding wall contact.

B. Contributions

The paper makes four contributions. First, we establish a sufficient condition for an initially feasible wall-distance HOCBF filter to lose feasibility before braking becomes available (Theorem 3).

Second, we derive an explicit stopping-distance certificate from a thrust-reserved swing-and-brake maneuver. It defines a controlled-invariant set with guaranteed filter feasibility under the stated continuous-time taut-cable assumptions (Theorem 9(i), Proposition 12).

Third, we derive an optimistic stopping distance that bounds the viability kernel from outside. It identifies states from which no admissible controller can avoid wall contact.

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Fourth, simulations evaluate the filter against HOCBF and numerical backup-CBF baselines. They quantify the conservatism of the certificate and the margin that a sampled implementation needs.

II. PROBLEM FORMULATION

Quadrotor $i \in \{1, \dots, N\}$ has position $p_i \in \mathbb{R}^3$, mass m_i , and thrust vector $u_i := f_i R_i e_3$, where $f_i \geq 0$ is the thrust magnitude, $R_i \in SO(3)$ the attitude, and e_3 the vertical unit vector; an attitude controller produces the commanded thrust vectors (Remark 14). Each thrust vector lies in the ball $\mathcal{F}_i := \{u \in \mathbb{R}^3 : \|u\| \leq f_{\max,i}\}$.

The payload is a point mass m_L at $x_L \in \mathbb{R}^3$, attached to quadrotor i by a massless, inextensible cable of length l_i with unit direction $q_i \in \mathbb{S}^2$ from the payload to the quadrotor, $p_i = x_L + l_i q_i$. The cable exerts the force $T_i q_i$ on the payload and $-T_i q_i$ on the quadrotor, with $T_i \geq 0$ because a cable cannot push. With the gravitational acceleration g ,

$$m_L \ddot{x}_L = \sum_i T_i q_i - m_L g e_3, \quad (1)$$

$$m_i \ddot{p}_i = u_i - m_i g e_3 - T_i q_i. \quad (2)$$

The state is $x := (x_L, \dot{x}_L, q, \dot{q})$ with $q := (q_1, \dots, q_N)$, where $q_i^\top q_i = 1$ and $\dot{q}_i^\top q_i = 0$. We assume every cable is taut, with tension at least $T_{\min} > 0$; the filter in Sec. V enforces this bound, and the simulations model slack cables.

We decompose each thrust vector along the cable and perpendicular to it, $u_i = s_i q_i + u_i^\perp$ with $s_i := q_i^\top u_i$, $u_i^\perp := P_i u_i$, $P_i := I - q_i q_i^\top$, and we use the *specific force* of the payload, $a := \ddot{x}_L + g e_3$.

Lemma 1 (Structure). *Along any solution of (1) with taut cables: (a) $T_i = s_i - m_i q_i^\top a + m_i l_i \|\dot{q}_i\|^2$ with $s = (s_1, \dots, s_N)$; (b) $\ddot{q}_i = \frac{1}{m_i l_i} u_i^\perp - \frac{1}{l_i} P_i a - \|\dot{q}_i\|^2 q_i$; (c) $M_a(q) a = \sum_i s_i q_i + \sum_i m_i l_i \|\dot{q}_i\|^2 q_i$ with $M_a(q) := m_L I + \sum_i m_i q_i q_i^\top \succ 0$.*

Proof. Differentiate $p_i = x_L + l_i q_i$ twice and substitute (1). Since $q_i^\top \dot{q}_i = 0$ and $\dot{q}_i^\top \ddot{q}_i = -\|\dot{q}_i\|^2$, the projection along q_i gives (a) and the projection perpendicular to q_i gives (b); substituting (a) into (1) gives (c). $\square$

In words, the parallel component sets the tension, the perpendicular component swings the cable, the two share the limit $\|s_i q_i + u_i^\perp\| \leq f_{\max,i}$, and the quadrotors interact only through the specific force, which by (c) the inputs determine uniquely. For fixed $(q, \dot{q})$, (a) and (c) define an affine bijection between s and $T = (T_1, \dots, T_N)$, so we use the tensions as decision variables: given T , $a = \frac{1}{m_L} \sum_j T_j q_j$, and (a) returns $s_i(T)$.

We consider a wall with horizontal unit normal n pointing into it, with the constraint

$$h(x_L) := d_0 - n^\top x_L \geq 0, \quad (3)$$

where h is the distance from the payload to the wall. We write $v := n^\top \dot{x}_L$ for the approach speed, $y := -n$ for the braking direction, and $b := y^\top \ddot{x}_L$ for the braking deceleration, so that $\dot{h} = -v$ and $\dot{v} = -b$, and we describe each cable direction by the *swing coordinates* $z_i := q_i^\top y$ and $w_i := q_i^\top r$, $r := e_3 \times y$. A cable with $z_i > 0$ lies in the *braking half-space*: it leans

away from the wall and pulls the payload away from it; a cable with $z_i < 0$ pulls the payload toward the wall.

We impose the operational constraints

$$\mathcal{X}_{\text{op}} := \{x : e_3^\top q_i \geq \cos \theta_q, |z_i| \leq \bar{z}, (w_i, \dot{w}_i) \in \mathcal{V}(\nu_w, \bar{w}), \|\dot{q}_i\| \leq \bar{\omega} \forall i\}, \quad (4)$$

with $\bar{z}^2 + \bar{w}^2 \leq \sin^2 \theta_q$, $\theta_q < \pi/2$. The upper-cone condition $e_3^\top q_i \geq \cos \theta_q$, together with the bounds on (z_i, w_i) , keeps every cable within θ_q of the upward vertical; the remaining conditions reserve stopping room for the transverse swing and bound the swing rate. The set $\mathcal{V}$ is defined in (10) below. With the altitude $\chi := e_3^\top x_L$ of the payload and a band $[\chi_{\min}, \chi_{\max}]$, the constraint set is $\mathcal{C} := \{x \in \mathcal{X}_{\text{op}} : h \geq 0, \chi_{\min} \leq \chi \leq \chi_{\max}\}$. The inputs must respect the thrust limits, the tension bound, and a bound $a_{\max}$ on the payload's specific force:

$$\mathcal{U}(x) := \{u : u_i \in \mathcal{F}_i, T_i(x, u) \geq T_{\min} \forall i, \|\sum_i T_i(x, u) q_i\| \leq m_L a_{\max}\}. \quad (5)$$

The last constraint is a design limit on the payload's acceleration; the bound that follows from the thrust limits alone is too large for Assumption 4 to hold with realistic constants.

A state is *viable* if some input signal with values in $\mathcal{U}(x)$ keeps the trajectory in $\mathcal{C}$ forever; the set of viable states is the *viability kernel* $\text{Viab}(\mathcal{C})$ [17], outside of which no controller can keep the system safe. A set is *controlled invariant* if some admissible input signal keeps every trajectory that starts in it inside it; every controlled-invariant subset of $\mathcal{C}$ lies in $\text{Viab}(\mathcal{C})$, which has no closed form. The problem is: (a) find a controlled-invariant set $\mathcal{S} \subseteq \mathcal{C}$ whose membership can be tested online; (b) find an explicit set that contains $\text{Viab}(\mathcal{C})$; and (c) derive a safety filter that is feasible at every state of $\mathcal{S}$.

III. LOSS OF FEASIBILITY OF HIGH-ORDER CBF FILTERS

For a unit vector y , the *exact directional authority* at x ,

$$\alpha_y^{\text{ex}}(x) := \max\{y^\top \ddot{x}_L(x, u) : u \in \mathcal{U}(x)\}, \quad (6)$$

is the largest acceleration that admissible inputs can give the payload along y ; the maximum exists because $\ddot{x}_L$ is affine in u and $\mathcal{U}(x)$ is compact and convex ($-\infty$ if empty). Since it is attained, for any function β of the state,

$$\begin{aligned} \{u \in \mathcal{U}(x) : b(x, u) \geq \beta(x)\} &\neq \emptyset \\ \iff \alpha_y^{\text{ex}}(x) &\geq \beta(x). \end{aligned} \quad (7)$$

Since h has relative degree two with respect to the tensions, HOCBFs [1], [3] apply the barrier condition twice. Let α_1 be a continuously differentiable class- $\mathcal{K}$ function, α_2 an extended class- $\mathcal{K}$ function, and $\psi_1 := \dot{h} + \alpha_1(h) = -v + \alpha_1(h)$. With $\dot{v} = -b$, the HOCBF condition $\dot{\psi}_1 \geq -\alpha_2(\psi_1)$ reads

$$b \geq \beta(x) := \alpha_1'(h) v - \alpha_2(\psi_1). \quad (8)$$

The HOCBF filter minimizes the change to the nominal command over $K_{\text{HO}}(x) := \{u \in \mathcal{U}(x) : b(x, u) \geq \beta(x)\}$ and, while $K_{\text{HO}} \neq \emptyset$, keeps the state in $\mathcal{C}_{\text{HO}} := \{h \geq$

0, $\psi_1 \geq 0$ [2]. By (7), $K_{\text{HO}}(x) \neq \emptyset$ if and only if the feasibility margin $\mu(x) := \alpha_y^{\text{ex}}(x) - \beta(x)$ is nonnegative.

Definition 2 (No-braking set). *The no-braking set for the braking direction y is $\Sigma_y := \{q \in (\mathbb{S}^2)^N : z_i \leq 0, i = 1, \dots, N\}$.*

In Σ_y no cable lies in the braking half-space, so no admissible input can decelerate the payload; the set has nonempty interior and contains the configurations of a team accelerating toward the wall.

Theorem 3 (Conditional loss of HOCBF feasibility). *Let $x_0 \in \mathcal{C}_{\text{HO}} \cap \mathcal{X}_{\text{op}}$ have $z_i(0) < 0$ for all i , $0 < v_0 < \alpha_1(h_0)$, and $\mu(x_0) > 0$. Let $\tau := \min_i(-z_i(0))/\bar{\omega}$ and $t_\psi := [h_0 - \alpha_1^{-1}(v_0)]/v_0$. If $t_\psi < \tau$, then every admissible trajectory that remains in $\mathcal{X}_{\text{op}}$ has a nonempty interval before τ with $h > 0$, $q \in \Sigma_y$, and $K_{\text{HO}} = \emptyset$. The property persists on an open neighborhood of x_0 if the input set varies continuously and stays nonempty there.*

Proof. Until τ , $z_i(t) \leq z_i(0) + \bar{\omega}t \leq 0$. Because $T_i \geq T_{\min}$, $b = m_L^{-1} \sum_i T_i z_i \leq 0$, hence $v(t) \geq v_0$ and $h(t) \leq h_0 - v_0 t$. The first time h reaches $\alpha_1^{-1}(v_0) > 0$ occurs by $t_\psi < \tau$. Immediately afterwards $\psi_1 < 0$ while $h > 0$. Since $\alpha_1'(h) \geq 0$ and α_2 is extended class- $\mathcal{K}$, $\mu = \alpha_y^{\text{ex}} - \alpha_1'(h)v + \alpha_2(\psi_1) < 0$. The inequalities are strict at the initial state and the authority maximum is continuous under the stated regularity, so they persist locally. $\square$

The initial-feasibility hypothesis cannot be removed: with $\alpha_1(h) = kh$ and ψ_1 near zero the required braking is close to $kv > 0$, which Σ_y cannot provide, so such states already have an infeasible QP, and a claim for every choice of gains would be too strong. For the two quadrotors of Fig. 1 with $m_L = m_i = l_i = 1$ in SI units, $T_{\min} = 1$ N, $\bar{\omega} = 2$ s⁻¹, $\alpha_1(h) = h$, $\alpha_2(s) = 20s$, $h_0 = 1$ m, and $v_0 = 0.85$ m/s, the floor tensions give $b = -1$ m/s², the HOCBF threshold is $\beta = -2.15$ m/s², $t_\psi = 0.176$ s, and $\tau = 0.25$ s; provided those tensions satisfy the thrust constraints, this is an initially feasible instance that loses feasibility.

IV. THE AUTHORITY BARRIER

To obtain a braking authority that depends only on the cable directions, we fix in advance how each quadrotor divides its thrust between tension and swinging. Throughout, $(z)_+ := \max\{0, z\}$ and $(z)_- := \max\{0, -z\}$.

Assumption 4 (Thrust reserve). *There exist constants $\bar{T}_i \geq T_{\min}$ (tension limit) and $\rho_i \geq 0$ (swing reserve) with $\sum_j \bar{T}_j \leq m_L a_{\max}$ such that, for every $x \in \mathcal{X}_{\text{op}}$, every T with $T_{\min} \leq T_j \leq \bar{T}_j$ for all j , and every $u_i^\perp \perp q_i$ with $\|u_i^\perp\| \leq \rho_i$, the thrust vector $s_i(T)q_i + u_i^\perp$ lies in $\mathcal{F}_i$.*

In words, each quadrotor can produce any tension up to $\bar{T}_i$ and, simultaneously, any perpendicular thrust up to ρ_i , and such tensions respect the cone in (5); by Lemma 1(a), the assumption holds if $(\bar{T}_i + m_i a_{\max} + m_i l_i \bar{\omega}^2)^2 + \rho_i^2 \leq f_{\max, i}^2$.

Definition 5 (Directional authority). *Let the hover floors $T_i^h \in [T_{\min}, \bar{T}_i]$ satisfy $\cos \theta_q \sum_i T_i^h \geq m_L g$, and let $\phi_i(z) :=$*

$\bar{T}_i(z)_+ - T_i^h(z)_-$. *The directional authority of q along the braking direction y is*

$$\alpha_y(q) := \frac{1}{m_L} \sum_{i=1}^N \phi_i(z_i), \quad z_i = q_i^\top y. \quad (9)$$

A cable in the braking half-space contributes most at $\bar{T}_i$, a cable that leans toward the wall opposes least at its floor; each ϕ_i is nondecreasing, convex, and piecewise linear with one breakpoint. The floor is T_i^h and not $T_{\min}$ because tensions at the floors carry the payload for every cable configuration of $\mathcal{X}_{\text{op}}$: a cable that leans toward the wall and carries its share of the weight pushes the payload toward the wall, and the authority charges this push.

Lemma 6 (Authority as a support function). *The set $\mathcal{Z}(q) := \{\sum_i T_i q_i : T_i^h \leq T_i \leq \bar{T}_i\}$ of tension forces is a zonotope, and $m_L \alpha_y(q) = \max_{f \in \mathcal{Z}(q)} y^\top f$ is its support function along y . Under Assumption 4, every $b \in [-\alpha_{-y}(q), \alpha_y(q)]$ is produced by some input in $\mathcal{U}(x)$ at every $x \in \mathcal{X}_{\text{op}}$; hence $\alpha_y \leq \alpha_y^{\text{ex}}$.*

Proof. $\mathcal{Z}(q)$ is the Minkowski sum of the segments $[T_i^h, \bar{T}_i] q_i$, the support function of a Minkowski sum is the sum of the support functions, and $\max_{T \in [T_i^h, \bar{T}_i]} T z_i = \phi_i(z_i)$. Assumption 4 with $u_i^\perp = 0$ realizes every T in the box, and $b = \frac{1}{m_L} \sum_i T_i z_i$ is affine in T . $\square$

A. Systems with second-order authority

We state the construction for a general system with coordinates $\zeta_1, \dots, \zeta_N$ that bound the braking authority monotonically and that the inputs drive only through their second derivatives. For $\nu, \bar{\zeta} > 0$, the admissible swing set

$$\mathcal{V}(\nu, \bar{\zeta}) := \{(\zeta, \dot{\zeta}) \in \mathbb{R}^2 : -\bar{\zeta} + \frac{(\omega)_-^2}{2\nu} \leq \zeta \leq \bar{\zeta} - \frac{(\omega)_+^2}{2\nu}\} \quad (10)$$

is the convex set of $(\zeta, \dot{\zeta})$ from which a double integrator with $|\ddot{\zeta}| \leq \nu$ can keep $|\zeta| \leq \bar{\zeta}$ for all future time.

Definition 7 (Second-order authority). *Consider a system with admissible inputs, a constraint function h with $\dot{h} = -v$ and $\ddot{h} = -b$, a closed operational set $\mathcal{X}_{\text{op}}$, and twice continuously differentiable coordinates $\zeta_1, \dots, \zeta_N$ of the state. The system has second-order authority along h on $\mathcal{X}_{\text{op}}$ with data $(\underline{\alpha}, \bar{\alpha}, \nu, \nu^{\text{rel}}, \bar{\zeta})$ if:*

- (C1) $\underline{\alpha}$ is the sum of a constant and of N continuous, non-decreasing, piecewise-linear functions of the individual ζ_i ; $\bar{\alpha} \geq \underline{\alpha}$ is continuous and nondecreasing in each ζ_i ; and $b \leq \bar{\alpha}(\zeta)$ for every admissible input at every $x \in \mathcal{X}_{\text{op}}$.
- (C2) $|\zeta_i| \leq \bar{\zeta}$ on $\mathcal{X}_{\text{op}}$, and $|\ddot{\zeta}_i| \leq \nu^{\text{rel}}$ for every admissible input at every $x \in \mathcal{X}_{\text{op}}$.
- (C3) There is a map that assigns to every $x \in \mathcal{X}_{\text{op}}$ with $(\zeta_i, \dot{\zeta}_i) \in \mathcal{V}(\nu, \bar{\zeta})$ for all i , and to every choice of $\ddot{\zeta}_i^d \in [-\nu, \nu]$, an admissible input that produces $b = \bar{\alpha}(\zeta)$ and $\ddot{\zeta}_i = \ddot{\zeta}_i^d$ for all i , and whose solutions remain in $\mathcal{X}_{\text{op}}$ as long as $(\zeta_i, \dot{\zeta}_i) \in \mathcal{V}(\nu, \bar{\zeta})$ for all i .

In words, no admissible input brakes harder than $\bar{\alpha}$ or swings faster than ν^{rel} , while a reserved part of the input

swings every coordinate at any rate up to ν and brakes with $\underline{\alpha}$ at the same time. For such a system the wall constraint alone defines $\mathcal{C}_h := \{x \in \mathcal{X}_{\text{op}} : h \geq 0\}$.

B. The braking maneuver and its stopping distance

Fix a deceleration $\nu' \in [\nu/2, \nu]$. For $(\zeta, \omega) \in \mathcal{V}(\nu, \bar{\zeta})$, let $\tilde{\zeta}(t; \zeta, \omega)$ be the motion from (ζ, ω) to $(\bar{\zeta}, 0)$ that accelerates at ν and decelerates at ν' , held at $\bar{\zeta}$ afterwards: with $r := \sqrt{2\nu\nu'(\bar{\zeta} - \zeta) + \nu'\omega^2}/(\nu + \nu')$, $t_1 := (r - \omega)/\nu$, and $t_2 := t_1 + r/\nu'$,

$$\tilde{\zeta}(t) = \begin{cases} \zeta + \omega t + \frac{\nu}{2}t^2, & 0 \leq t \leq t_1, \\ \bar{\zeta} - \frac{\nu'}{2}(t_2 - t)^2, & t_1 \leq t \leq t_2, \\ \bar{\zeta}, & t \geq t_2, \end{cases} \quad (11)$$

if $\omega \leq 0$ or $\omega^2 < 2\nu'(\bar{\zeta} - \zeta)$; otherwise, the motion decelerates from the start at the constant rate $\omega^2/(2(\bar{\zeta} - \zeta)) \in [\nu', \nu]$ and arrives at $t_2 = 2(\bar{\zeta} - \zeta)/\omega$. Because $\nu' < \nu$, the swing arrives through the interior of $\mathcal{V}(\nu, \bar{\zeta})$; the time-optimal swing, $\nu' = \nu$, is denoted $\tilde{\zeta}^\circ$. The swing is continuously differentiable in its initial data except on the curve $\omega^2 = 2\nu'(\bar{\zeta} - \zeta)$, $\omega \geq 0$, where its two cases meet, and its derivatives are one-sided. A state is *regular* if none of its swing states lies on this curve, and *degenerate* otherwise.

Let x have $(\zeta_i, \dot{\zeta}_i) \in \mathcal{V}(\nu, \bar{\zeta})$ for all i , and let every coordinate follow its swing $\tilde{\zeta}_i(t) := \tilde{\zeta}(t; \zeta_i, \dot{\zeta}_i)$. Along the maneuver, the braking deceleration is $\tilde{\alpha}(t; x) := \underline{\alpha}(\tilde{\zeta}_1(t), \dots, \tilde{\zeta}_N(t))$, the approach speed $\tilde{v}(t; x) := v - \int_0^t \tilde{\alpha}(s; x) ds$, and the distance traveled toward the wall $E(t; x) := \int_0^t \tilde{v}(s; x) ds$. The *stopping distance* is

$$D(x) := \max_{t \geq 0} E(t; x), \quad (12)$$

with maximizing times $T^*(x)$; the maximum exists if $\underline{\alpha}_{\text{sat}} := \underline{\alpha}(\bar{\zeta}, \dots, \bar{\zeta}) > 0$, and it is taken over all $t \geq 0$, not up to the first zero of $\tilde{v}$, because a negative authority accelerates a payload at rest toward the wall. The *authority barrier* is $H(x) := h - D(x)$, and

$$\mathcal{X}_{\text{RF}} := \{x \in \mathcal{X}_{\text{op}} : H(x) \geq 0, (\zeta_i, \dot{\zeta}_i) \in \mathcal{V}(\nu, \bar{\zeta}) \forall i\}. \quad (13)$$

$H(x)$ is the smallest distance to the wall that the payload reaches during the maneuver started at x . The functions $h_i^+ := \bar{\zeta} - \zeta_i - (\dot{\zeta}_i^2)/(2\nu)$ and $h_i^- := \bar{\zeta} + \zeta_i - (\dot{\zeta}_i^2)/(2\nu)$ are nonnegative exactly on $\mathcal{V}(\nu, \bar{\zeta})$. For the swing-set barrier condition below, define the signed square root $\text{sroot}(s) := \text{sgn}(s)\sqrt{|s|}$, with $\text{sroot}(0) := 0$; on $s \geq 0$ this is the ordinary square root. Let $\pi^\sharp(x)$ be the input that (C3) assigns to x with $\tilde{\zeta}_i^\sharp = \tilde{\zeta}(0; \zeta_i, \dot{\zeta}_i)$; since the swing is a feedback of $(\zeta_i, \dot{\zeta}_i)$, the closed loop under $\pi^\sharp$ reproduces the swings, $b(t) = \tilde{\alpha}(t; x_0)$, and $v(t) = \tilde{v}(t; x_0)$.

Lemma 8 (Envelope and regularity). (i) Every $\zeta(\cdot)$ with absolutely continuous derivative, $\zeta(0) = \zeta$, $\dot{\zeta}(0) = \omega$, $|\dot{\zeta}| \leq \nu$ a.e., and $\zeta \leq \bar{\zeta}$ satisfies $\zeta(t) \leq \tilde{\zeta}^\circ(t; \zeta, \omega)$ for all $t \geq 0$. (ii) D is locally Lipschitz on the states with $(\zeta_i, \dot{\zeta}_i) \in \mathcal{V}(\nu, \bar{\zeta})$ for all i , and along the maneuver started at x_0 , $D(x(t)) = \max_{s \geq t} \int_0^s \tilde{v}(\cdot; x_0) - \int_0^t \tilde{v}(\cdot; x_0)$.

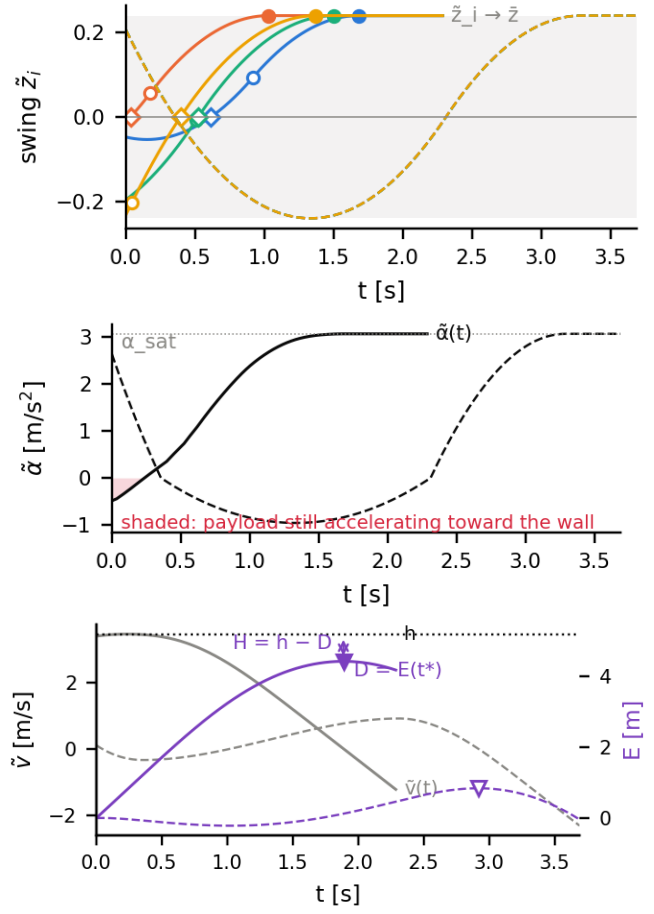


Fig. 2. The reserved swing (top), the braking authority it releases (middle), and the wallward excursion $E(t)$, whose maximum is D (bottom), for two states (solid, dashed).

Proof scheme. (i) Write $S(\zeta) := \sqrt{2\nu(\bar{\zeta} - \zeta)}$. A trajectory moving toward $\bar{\zeta}$ must satisfy $\dot{\zeta} \leq S(\zeta)$, or even full braking would overshoot. Before its switch, $\tilde{\zeta}^\circ$ uses the largest acceleration ν ; afterwards it follows $\tilde{\zeta}^\circ = S(\tilde{\zeta}^\circ)$ to $\bar{\zeta}$. A first crossing from below either contradicts maximal acceleration before the switch or forces both motions to coincide on the stopping curve after it. (ii) On compact initial-state sets the arrival times and the maximizing times are uniformly bounded because $\underline{\alpha}_{\text{sat}} > 0$, and $E(t; \cdot)$ is locally Lipschitz uniformly in t ; the maximum of such a family is locally Lipschitz. Restarting the swing at time t reproduces its remaining trajectory, which gives the time-shift identity. $\square$

C. Main result

At regular states, for fixed t , $E(t; x)$ is continuously differentiable in $(v, \zeta, \dot{\zeta})$, because the integration over s smooths the breakpoints of $\underline{\alpha}$, which each $\tilde{\zeta}_i$ meets only at isolated times except while held at $\bar{\zeta}$. There, Danskin's theorem [18] gives the directional derivative

$$D'(x; \delta) = \max_{t \in T^*(x)} \nabla_x E(t; x)^\top \delta \quad (14)$$

in directions δ along which the state stays in the domain of D . For an admissible u , let $F_u(x)$ be the state derivative and let $H'(x; F_u(x))$ be the upper right Dini derivative of H along

it; at regular states it equals $-v - D'(x; F_u(x))$, the ordinary derivative $\dot{H}(x, u)$ when the maximizing time is unique. On states with all $(\zeta_i, \dot{\zeta}_i) \in \mathcal{V}(\nu^{\text{rel}}, \bar{\zeta})$, let D^{rel} be (12) computed with $(\bar{\alpha}, \nu^{\text{rel}})$ in place of $(\underline{\alpha}, \nu)$ and with the time-optimal swing: the stopping distance of an optimistic system that swings and brakes as hard as any admissible input allows. Let γ be a locally Lipschitz extended class- $\mathcal{K}$ function.

Theorem 9 (Inner and outer bounds). *Let the system have second-order authority along h on $\mathcal{X}_{\text{op}}$ with data $(\underline{\alpha}, \bar{\alpha}, \nu, \nu^{\text{rel}}, \bar{\zeta})$, and let $\underline{\alpha}_{\text{sat}} > 0$. Then:*

- (i) $\mathcal{X}_{\text{RF}}$ is controlled invariant. For every $x \in \mathcal{X}_{\text{RF}}$, the set $K_H(x)$ of the admissible u with $H'(x; F_u(x)) \geq -\gamma(H(x))$ and $\dot{h}_i^\pm \geq -c_V \text{sroot}(h_i^\pm)$ for all i , $c_V := 2\sqrt{2\nu\nu'}/(\nu - \nu')$, contains $\pi^\sharp(x)$. Every solution in $\mathcal{X}_{\text{op}}$ generated by a measurable selection $u(t) \in K_H(x(t))$ from $x(0) \in \mathcal{X}_{\text{RF}}$ satisfies $(\zeta_i, \dot{\zeta}_i) \in \mathcal{V}(\nu, \bar{\zeta})$ and $h(t) \geq H(x(t)) \geq 0$ for all $t \geq 0$.
- (ii) $\text{Viab}(\mathcal{C}_h) \subseteq \{x \in \mathcal{X}_{\text{op}} : (\zeta_i, \dot{\zeta}_i) \in \mathcal{V}(\nu^{\text{rel}}, \bar{\zeta}) \forall i, h \geq D^{\text{rel}}(x)\}$.

Proof. (i) Under $\pi^\sharp$ from $x_0 \in \mathcal{X}_{\text{RF}}$, Lemma 8(ii) and $h(t) = h_0 - \int_0^t v$ give $H(x(t)) = h_0 - \max_{s \geq t} \int_0^s v$, nondecreasing in t ; $\pi^\sharp$ keeps the state in $\mathcal{X}_{\text{op}}$ and the swing states in $\mathcal{V}$, so $\mathcal{X}_{\text{RF}}$ is forward invariant under $\pi^\sharp$, hence controlled invariant. Next, the swing restarted at a later state of the maneuver is the remaining part of the swing started at x , so the derivative of $E(t; \cdot)$ along $F_{\pi^\sharp}(x)$ equals $\tilde{v}(t; x) - v$ at regular states, and by (14), $H'(x; F_{\pi^\sharp}(x)) = -\max_{t \in T^*(x)} \tilde{v}(t; x)$. Every maximizing time $t > 0$ is a stationary point of $E(\cdot; x)$, so $\tilde{v}(t; x) = 0$, and $t = 0$ maximizes only if $v \leq 0$; hence $H'(x; F_{\pi^\sharp}(x)) \geq 0 \geq -\gamma(H(x))$ at regular states. The time-shift identity gives the same inequality for the Dini derivative along $\pi^\sharp$ at degenerate states. The swing of $\pi^\sharp$ satisfies the conditions on $\dot{h}_i^\pm$: $\dot{h}_i^- \geq 0$; while it accelerates with $\dot{\zeta}_i > 0$, $\dot{h}_i^+ = -2\dot{\zeta}_i$ and $\dot{\zeta}_i^2 < 2\nu\nu'(\bar{\zeta} - \zeta_i) < 2\nu\nu'\dot{h}_i^+ / (\nu - \nu')$; while it decelerates at a rate $a \in [\nu', \nu]$, $(\dot{h}_i^+)^2 = 2a(1 - a/\nu)\dot{h}_i^+ \leq 2(\nu - \nu')\dot{h}_i^+ \leq c_V^2 \dot{h}_i^+$. Finally, along a solution generated by K_H no $\dot{h}_i^\pm$ becomes negative: on an interval where it is negative, starting at a zero, its derivative is at least $c_V \sqrt{-\dot{h}_i^\pm} > 0$ a.e., which contradicts absolute continuity. Thus the swing states stay in $\mathcal{V}$, where D is locally Lipschitz, so $H(x(t))$ is absolutely continuous with $H'(x; \dot{x}) \geq -\gamma(H)$ a.e., and the comparison argument for nonsmooth barriers [19] gives $H \geq 0$; $h \geq H$ since $D \geq 0$.

(ii) Let $x_0 \in \text{Viab}(\mathcal{C}_h)$ and let an admissible input keep the solution in $\mathcal{C}_h$. By (C2), $(\zeta_i, \dot{\zeta}_i) \in \mathcal{V}(\nu^{\text{rel}}, \bar{\zeta})$ at all times, and by Lemma 8(i) with ν^{rel} , $\zeta_i(t) \leq \bar{\zeta}_i^{\text{rel}}(t)$. By (C1) and the monotonicity of $\bar{\alpha}$, $b(t) \leq \bar{\alpha}(\zeta(t)) \leq \bar{\alpha}(\bar{\zeta}^{\text{rel}}(t)) = \bar{\alpha}^{\text{rel}}(t; x_0)$, so $v(t) \geq \bar{v}^{\text{rel}}(t; x_0)$ and $\min_t h(t) \leq h_0 - D^{\text{rel}}(x_0)$; if $h_0 < D^{\text{rel}}(x_0)$, the solution leaves $\mathcal{C}_h$. $\square$

Together, (i) and (ii) give $\mathcal{X}_{\text{RF}} \subseteq \text{Viab}(\mathcal{C}_h) \subseteq \{h \geq D^{\text{rel}}\}$. The gap $D - D^{\text{rel}}$ is what the division of the input costs; for a constant authority $\underline{\alpha} \equiv \bar{\alpha} \equiv \bar{\alpha} > 0$ both distances equal $(v)_+^2 / (2\bar{\alpha})$, the braking-distance barrier of [20]. Part (i) is a backup-CBF guarantee [6], [7] with $\pi^\sharp$ as the backup controller, but the closest approach along the backup trajectory

has the closed form (12) instead of being simulated, and the same data yield the outer bound (ii). The sets $\mathcal{X}_{\text{RF}}$ and $\mathcal{C}_{\text{HO}}$ need not be nested.

D. Application to the cable system

Assumption 10 (Swing reserve). *There exist $\nu, \nu_w > 0$ and $\rho_i^{\text{fb}} \geq 0$ such that, for every i , $\sec \theta_q \sqrt{\bar{\eta}_{z,i}^2 + \bar{\eta}_{w,i}^2} + \rho_i^{\text{fb}} \leq \rho_i$, where $\bar{\eta}_{z,i} := m_i(l_i \nu + a_{\max} + l_i \bar{\omega}^2 \bar{z})$ and $\bar{\eta}_{w,i} := m_i(l_i \nu_w + a_{\max} + l_i \bar{\omega}^2 \bar{w})$.*

Assumption 11 (Swing-rate consistency). $\bar{\omega}^2 \geq \sec^2 \theta_q (4\nu \bar{z} + 4\nu_w \bar{w})$.

Assumption 10 reserves, within ρ_i , the perpendicular thrust needed to impose any $\ddot{z}_i \in [-\nu, \nu]$ and $\ddot{w}_i \in [-\nu_w, \nu_w]$ against the coupling terms of Lemma 1(b), the factor $\sec \theta_q$ accounting for the non-orthogonality of the two swing directions on the sphere, and keeps ρ_i^{fb} for feedback; Assumption 11 makes the swing-rate bound of (4) hold on the admissible swing sets.

The maneuver also bounds the altitude. Its tensions lie in $[T_i^h, \bar{T}_i]$, so the vertical acceleration $\ddot{\chi} = \frac{1}{m_L} \sum_i T_i e_3^\top q_i - g$ lies between $c_0 := \frac{\cos \theta_q}{m_L} \sum_i T_i^h - g \geq 0$ and $c_1 := \frac{1}{m_L} \sum_i \bar{T}_i - g$ until the stopping time $t_s(x) := \max\{t_a + \bar{v}(t_a; x)/\underline{\alpha}_{\text{sat}}, t_a - \epsilon, 0\}$, $t_a := \max_i t_{2,i}$, with a lead $\epsilon > 0$ small enough that $z_i > 0$ whenever $t_{2,i} \leq \epsilon$; at t_s every swing is about to arrive and the payload recedes from the wall. From then on $\pi^\sharp$ holds the altitude with one uniform tension that decelerates a climb at some $c_\downarrow > 0$ satisfying $c_\downarrow \leq g - \frac{N}{m_L} T_{\min}$ and a descent at some $c_\uparrow > 0$ satisfying $c_\uparrow \leq \frac{\cos \theta_q}{m_L} \sum_i \bar{T}_i - g$. With $U := \dot{\chi} + c_1 t_s$ and $L := \dot{\chi} + c_0 t_s$, let $\Delta_\uparrow := (\dot{\chi} t_s + \frac{c_1}{2} t_s^2 + \frac{U^2}{2c_\downarrow})_+$ if $U > 0$ and $\Delta_\uparrow := 0$ otherwise; let $\Delta_\downarrow := 0$ if $\dot{\chi} \geq 0$, $\Delta_\downarrow := \dot{\chi}^2 / (2c_0)$ if $\dot{\chi} < 0 \leq L$, and $\Delta_\downarrow := -\dot{\chi} t_s - \frac{c_0}{2} t_s^2 + \frac{L^2}{2c_\uparrow}$ if $L < 0$. The altitude barriers are

$$H_\uparrow := \chi_{\max} - \chi - \Delta_\uparrow, \quad H_\downarrow := \chi - \chi_{\min} - \Delta_\downarrow,$$

and the certified set is $\mathcal{S} := \mathcal{X}_{\text{RF}} \cap \{H_\uparrow \geq 0, H_\downarrow \geq 0\}$. At the switching surfaces of t_s , $\Delta_\uparrow$, and $\Delta_\downarrow$, $\dot{H}_\uparrow$ and $\dot{H}_\downarrow$ denote one-sided derivatives, and the inequalities below hold for every active branch.

Proposition 12 (The cable system has second-order authority). *Let Assumptions 4, 10, and 11 hold and let $\zeta_i := z_i$. Then (1)–(2) has second-order authority along h on $\mathcal{X}_{\text{op}}$ with $\underline{\alpha} = \alpha_y$ of (9),*

$$\bar{\alpha}(z) = \min \left\{ \frac{1}{m_L} \sum_i \phi_i^{\text{rel}}(z_i), a_{\max} \sec \theta_q \max_i (z_i)_+ \right\}$$

where $\phi_i^{\text{rel}}(z) := \bar{T}_i^{\text{rel}}(z)_+ - T_{\min}(z)_-$ with $\bar{T}_i^{\text{rel}} := f_{\max,i} + m_i a_{\max} + m_i l_i \bar{\omega}^2$ relaxes the tension limit upward and the floor down to $T_{\min}$, whereas $\underline{\alpha}$ keeps the hover floor; ν of Assumption 10; $\nu^{\text{rel}} := \max_i (\frac{f_{\max,i}}{m_i l_i} + \frac{a_{\max}}{l_i} + \bar{\omega}^2)$; and $\bar{\zeta} = \bar{z}$. The map of (C3) sets $T_i = \bar{T}_i$ if $z_i > 0$ and $T_i = T_i^h$ otherwise, and steers each w_i within $\mathcal{V}(\nu_w, \bar{w})$. Moreover, $\mathcal{S} \subseteq \mathcal{C}$ is controlled invariant, $\pi^\sharp$ satisfies $\dot{H}_\uparrow \geq 0$ and $\dot{H}_\downarrow \geq 0$, and Theorem 9(i) holds on $\mathcal{S}$ with these two conditions added to K_H .

Proof scheme. For (C1), every admissible input has $T_{\min} \leq T_i \leq T_i^{\text{rel}}$ by (5) and Lemma 1(a), and $T_i z_i \leq \phi_i^{\text{rel}}(z_i)$ for every such T_i , so $b = \frac{1}{m_L} \sum_i T_i z_i \leq \frac{1}{m_L} \sum_i \phi_i^{\text{rel}}(z_i)$. The upper-cone condition $e_3^\top q_i \geq \cos \theta_q$ in (4), together with the specific-force bound in (5), gives $\sum_i T_i \cos \theta_q \leq e_3^\top \sum_i T_i q_i \leq \|\sum_i T_i q_i\| \leq m_L a_{\max}$, hence also $b \leq a_{\max} \sec \theta_q \max_i (z_i)_+$. Projecting Lemma 1(b) onto y establishes (C2). For (C3), choose the tensions that realize $\underline{\alpha}$ and prescribe $\ddot{z}_i \in [-\nu, \nu]$ plus a transverse $\ddot{w}_i \in [-\nu_w, \nu_w]$ that keeps w_i in its admissible set. The Gram matrix of $(P_i y, P_i r)$ has eigenvalues 1 and $1 - \dot{z}_i^2 - \dot{w}_i^2 \geq \cos^2 \theta_q$, so the required perpendicular thrust has norm at most $\sec \theta_q \sqrt{\dot{\eta}_{z,i}^2 + \dot{\eta}_{w,i}^2} \leq \rho_i$, and Assumption 4 realizes the tensions and swings simultaneously. Finally, $\|\dot{q}_i\|^2 \leq \sec^2 \theta_q (\dot{z}_i^2 + \dot{w}_i^2) \leq \bar{\omega}^2$ on the product of the admissible swing sets, so the trajectory stays in $\mathcal{X}_{\text{op}}$. For the altitude, $\ddot{\chi} \in [c_0, c_1]$ while $\pi^\#$ brakes; comparison with the constant accelerations c_1 and c_0 until t_s , and with the decelerations of the hold afterward, bounds the climb by $\Delta_\uparrow$ and the drop by $\Delta_\downarrow$. Along $\pi^\#$, each positive argument of t_s decreases at unit rate, so on every branch $\dot{H}_\uparrow = (t_s + U/c_\downarrow)(c_1 - \ddot{\chi}) \geq 0$ where $\Delta_\uparrow > 0$, and likewise $\dot{H}_\downarrow \geq 0$. The hold keeps $b > 0$, so D is unchanged. $\square$

Hence $\mathcal{S} \subseteq \text{Viab}(\mathcal{C}) \subseteq \text{Viab}(\mathcal{C}_h) \subseteq \{h \geq D^{\text{rel}}\}$, which answers problems (a) and (b); Sec. V answers (c). With all cables at $\bar{z}$, $D^{\text{rel}} = (v_+^2)/(2a_{\max} \bar{z} \sec \theta_q)$ against $D = (v_+^2)/(2\alpha_{\text{sat}})$ with $\alpha_{\text{sat}} = \frac{1}{m_L} \sum_i \bar{T}_i \bar{z}$; where the cables must first swing, the gap is set by ν against ν^{rel} .

Proposition 13 (Computation). *For $t_j \in T^*(x)$, the time derivative of $h - E(t_j; \cdot)$ under u is*

$$\begin{aligned} \dot{H}_j(x, u) = & -v + t_j b \\ & - \sum_i (\partial_{z_i} E(t_j; x) \dot{z}_i + \partial_{\dot{z}_i} E(t_j; x) \ddot{z}_i), \end{aligned} \quad (15)$$

with $b = \frac{1}{m_L} \sum_i T_i z_i$ and $\ddot{z}_i = \frac{1}{m_i l_i} y^\top u_i^\perp - \frac{1}{l_i} y^\top P_i a - \|\dot{q}_i\|^2 z_i$, and $H'(x; F_u(x)) = \min_j \dot{H}_j(x, u)$ at regular states. D, T^* , and the coefficients of (15) depend on the state only through $(v, \{z_i, \dot{z}_i\}_i)$ and are computed exactly in $O(N \log N)$ arithmetic operations, because $E(\cdot; x)$ is piecewise quartic with at most $4N$ breakpoints; the gradient rows take a further $O(N|T^*|)$.

Proof scheme. At a regular state, $\partial_v E(t; x) = t$ and the chain rule give (15), and Danskin's formula makes the derivative of H the minimum over the active times. Every $\ddot{z}_i$ is piecewise quadratic with at most four breakpoints (switch, arrival, two zero crossings), so E is piecewise quartic, its candidate maximizers are interval endpoints or roots of the cubic $\ddot{v}$, and sorting the breakpoints costs $O(N \log N)$. $\square$

Remark 14 (Scope of the guarantee). The certificate assumes an exact taut-cable model, continuously updated thrust commands, a feasible transverse swing reserve, and the existence of closed-loop solutions. Wind, mass errors, actuator tracking, slack cables, and zero-order-hold inputs require additional error bounds and are *not* covered by Theorem 9.

In particular, the continuity and local Lipschitzness of D do not, by themselves, give a numerical sampled-data margin.

V. SAFETY FILTER

The decision variables are the tensions $T \in \mathbb{R}^N$ and the perpendicular components $u^\perp = (u_1^\perp, \dots, u_N^\perp)$, $u_i^\perp \perp q_i$; the thrust vector of quadrotor i is $s_i(T)q_i + u_i^\perp$ with $a = \frac{1}{m_L} \sum_j T_j q_j$ in Lemma 1(a). For each swing coordinate $\zeta \in \{z_i, w_i\}$ with data $(\nu_\zeta, \bar{\zeta}_\zeta) \in \{(\nu, \bar{z}), (\nu_w, \bar{w})\}$, let $h_\zeta^\pm$ be the two functions of Sec. IV-B that describe $\mathcal{V}(\nu_\zeta, \bar{\zeta}_\zeta)$, and let $c_a \geq 1$. Given the nominal thrust vectors u_i^{nom} , the *authority filter* solves

$$\min_{T, u^\perp} \sum_i \|s_i(T)q_i + u_i^\perp - u_i^{\text{nom}}\|^2 \quad (16a)$$

$$\text{s.t. } u \in \mathcal{U}(x), \quad (16b)$$

$$\dot{H}_j(x, T, u^\perp) \geq -\gamma(H(x)) \quad \forall t_j \in T^*(x), \quad (16c)$$

$$\dot{h}_\zeta^\pm \geq -c_V \text{sroot}(h_\zeta^\pm), \quad |\ddot{\zeta}| \leq c_a \nu_\zeta, \quad (16d)$$

$$\dot{H}_\uparrow \geq -\gamma(H_\uparrow), \quad \dot{H}_\downarrow \geq -\gamma(H_\downarrow), \quad (16e)$$

for every i and every swing coordinate. Constraint (16c) is the barrier condition of Theorem 9(i) written with Proposition 13, one affine constraint per maximizing time. The first condition of (16d) keeps the swing states in their admissible sets, with the signed root of Theorem 9(i), which is the ordinary square root on $\mathcal{S}$; it is affine in $\ddot{\zeta}$, and the swing of $\pi^\#$ satisfies it because it decelerates at $\nu' < \nu$, whereas the time-optimal swing reaches the boundary of $\mathcal{V}$ in finite time and satisfies no such condition. The second condition is the range of swing accelerations that $\pi^\#$ uses; it limits the change of the swing rates within one sampling period (Sec. VI). Constraint (16e) holds the altitude barriers of Sec. IV-D; their derivatives are affine in $(T, u^\perp)$ through $\ddot{\chi}$ and the derivative of t_s . At regular states, the program is a second-order cone program with $3N$ variables.

By Theorem 9(i) and Proposition 12, $\pi^\#(x)$ satisfies every constraint at the regular states of $\mathcal{S}$, with slack at least $\gamma(H)$ in (16c); at degenerate states the gradient rows are not asserted to characterize the Dini condition, and the filter applies $\pi^\#$ instead. Every closed-loop solution that starts in $\mathcal{S}$ stays in it, so $h \geq H \geq 0$ and $\chi_{\min} \leq \chi \leq \chi_{\max}$. The sampled implementation replaces the first condition of (16d) with the requirement that every swing state lies in its set $\mathcal{V}$ at the next sample, which is a second-order cone constraint.

VI. NUMERICAL STUDY

Figure 2 illustrates the predicted swing profiles. The team has four 1.5 kg quadrotors carrying a 1 kg point payload on 1 m cables, with $T_{\min} = 1$ N, $f_{\max} = 44$ N, $\bar{T}_i = 3.2$ N, $\rho_i = 34$ N, $a_{\max} = 13$ m/s², $\theta_q = 20^\circ$, $\bar{z} = 0.239$, $\bar{w} = 0.171$, $\bar{\omega} = 1$ s⁻¹, and $\nu = \nu_w = 0.5$ s⁻². The thrust-reserve bound holds: $\sqrt{(\bar{T}_i + m_i a_{\max} + m_i l_i \bar{\omega}^2)^2 + \rho_i^2} = 41.73$ N $< f_{\max}$. The code and trial videos are available online.¹ Throughout, $\gamma(s) = s$, the initial states are drawn with seed 11, the taut-cable model is integrated between updates with the

¹https://github.com/Hadi-Hajieghrary/Authority_Barriers

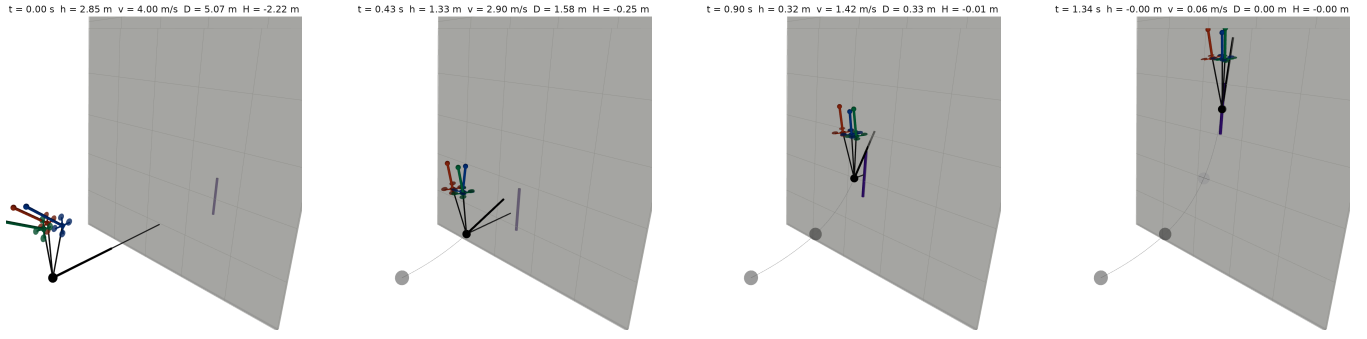


Fig. 3. A collocation trajectory for configuration B at $v_0 = 4$ m/s from $h_0 = 0.56 D$: the quadrotors swing their cables behind the payload and enter $\mathcal{X}_{\text{RF}}$.

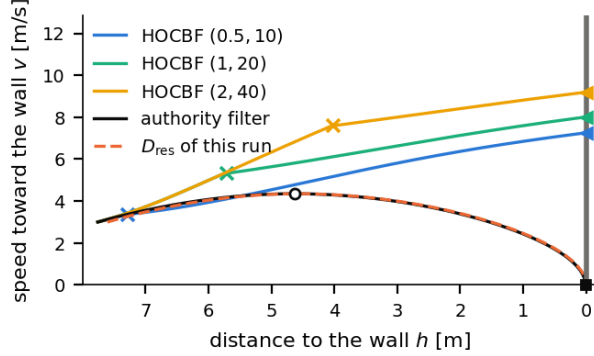


Fig. 4. Exact model, one state of $\mathcal{S}$, adversarial nominal input: wall distance against approach speed. The HOCBF filters lose feasibility (crosses) and reach the wall (triangles); the authority filter brings every cable into the braking half-space (circle) and stops (square). Dashed: D along its run.

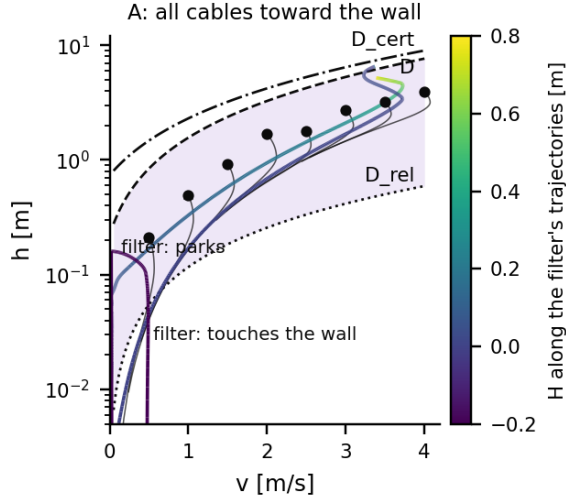


Fig. 5. Configuration A: stopping distances with the floor $T_{\min}$ (dashed), with the hover floor and v' (dash-dotted), and optimistic D^{rel} (dotted). Dots: smallest found collocation clearances. Colored curves: two runs of the sampled filter.

eighth-order Dormand–Prince method at relative and absolute tolerances 10^{-9} and 10^{-11} , and the programs are solved by Clarabel at its default tolerances through Drake 1.51.1 in Python 3.10.

Exact model. Theorem 9(i) was tested in its own setting: the authority filter (16) on the taut-cable model under a zero-order hold, with $T_i^h = 2.7$ N, $\nu' = 0.4$ s $^{-2}$, $c_{\downarrow} = 5$ and $c_{\uparrow} =$

2 m/s 2 , $\epsilon = 0.1$ s, $c_a = 1.25$, and the band $0 \leq \chi \leq 45$ m. The 100 initial states have random swing states, speeds of 1–4 m/s, and $H(x_0) \in [0, 0.05 D(x_0)]$, between 1 mm and 0.38 m; the nominal input is adversarial: the full thrust of every quadrotor toward the wall, tilted 40° upward; a trial lasts 6 s. Table I gives the result. With a period of 1 ms the barrier stayed above -0.05 mm, one trial reached the wall, by 2 μ m, and the altitude stayed in its band in every trial; with 0.5 ms no trial reached the wall and the barrier stayed above 0.22 mm; the payload climbed by up to 29.9 m.

The contact is a hold effect: rerun from the same state with the same nominal input, the trial stops 0.26 mm from the wall with a hold of 0.5 ms and 0.39 mm with 0.25 ms, whereas tightening the integrator tolerances to 10^{-12} or the solver tolerances to 10^{-10} at 1 ms changes the penetration by less than 1 nm; the three other states that reach the wall with 5 ms keep clearances of 0.27–1.7 mm at 1 ms and 0.31–2.3 mm at 0.25 ms. Between updates (ten states per hold) the minima of h , H , and the altitude barriers equal those at the updates to the printed digits, the swing states leave their sets by at most $2 \cdot 10^{-7}$, and every command respects the tension floor and the thrust limit. The certificate is continuous-time; its sampled implementation carries a margin that shrinks with the hold.

Figure 4 compares the authority filter with HOCBF filters with linear gains $(k_1, k_2) \in \{(0.5, 10), (1, 20), (2, 40)\}$ s $^{-1}$ from one state with $v = 3$ m/s, every cable at $z_i = -0.8\bar{z}$, and $h = 1.02 D = 7.76$ m. Each HOCBF program is feasible at the start and infeasible after 0.15, 0.52, and 0.78 s, with every cable leaning toward the wall; its relaxation with a penalized slack reaches the wall at 7.3–9.2 m/s. The authority filter has every cable in the braking half-space after 0.81 s and stops the payload 10 mm from the wall, with H between 0.15 and 0.01 m. The bound on the swing accelerations in (16d) is what keeps the hold benign: without it the filter follows the nominal input with swing accelerations of about 20 s $^{-2}$, forty times ν , and the stopping distance changes within one period by more than its gradient predicts; a margin of the kind derived for sampled-data CBFs [21] is not available for D . With the hover floor, the stopping distance from 3 m/s with every cable at $-\bar{z}$ is 9.18 m, against 6.56 m with the floor $T_{\min}$ and the time-optimal swing: carrying the weight costs 2.6 m.

TABLE I

AUTHORITY FILTER ON THE TAUT-CABLE MODEL, 100 TRIALS PER HOLD.

Period	Contacts	Deepest	min H	χ in band
0.5 ms	0	—	0.22 mm	100
1 ms	1	0.002 mm	−0.05 mm	100
5 ms	4	0.4 mm	−7 mm	100

The remaining trials use Full-order model simulations with rigid quadrotors, spring-damper cables that can go slack, thrust vectors applied directly or through a 1 kHz attitude loop, and a filter at 200 Hz with the wall barrier alone (the floor $T_{\min}$ in ϕ_i , the time-optimal swing, and neither (16e) nor the bound on the swing accelerations). All 690 HOCBF trials lost QP feasibility before a cable entered the braking half-space: the three gain pairs of Fig. 4, each from 100 states with every cable leaning toward the wall at 2–4 m/s under both actuation models, and from 30 states with three quadrotors; this does not establish initial feasibility for arbitrary gains. This model lies outside the hypotheses of Theorem 9: a cable that goes slack within a 5 ms hold can cost the barrier up to 0.85 m, so the sampled filter needs an empirical margin; in a sweep of 100 states under the adversarial nominal input, no trial started with $H(x_0) \geq 0.78$ m reached the wall. The margin holds only for the tested distribution.

Direct-collocation searches on the taut-cable model with three quadrotors (set A scaled to $N = 3$, $m_L = 0.75$ kg) covered three initial cable configurations at rest (A: $z_i = -0.15$; B: $z = (-0.15, 0, 0.15)$; C: $z_i = \bar{z}$) at eight speeds from 0.5 to 4 m/s, each candidate re-simulated. They impose every constraint of (4) and (5) with $h \geq 0$ and a terminal state in $\mathcal{X}_{\text{RF}}$, but no altitude band, and their D uses the floor $T_{\min}$ and the time-optimal swing; a search matched to the certified maneuver and band is left open. When cables already lean away from the wall, the found distance is 0.97–0.98 D, and the hover floor changes nothing there. When a swing is required, trajectories were found from 0.25–0.56 D at 15 of the 16 speeds (the search did not converge at the remaining one), which is 0.14–0.53 of the certified stopping distance with the hover floor and ν' (Fig. 5 shows configuration A with both). Figure 3 shows one such trajectory. The large gap is driven in part by the reserved swing acceleration: $\nu = 0.5 \text{ s}^{-2}$ needs 1.96 s to cross the allowed angle range, while the physical thrust can reach the swing-rate limit much sooner, and Assumption 11 admits no ν above 0.54 s^{-2} at this rate limit. A collocation success gives an example trajectory, but a failure does not locate the viability boundary, so the evidence supports the qualitative gap between the two models and validates neither set inclusion.

On one core of an Intel i7-1165G7, over 200 states of the exact-model sample, evaluating D with its gradients takes 0.7 ms (median; 95th percentile 0.8, maximum 1.0 ms), the filter with the wall barrier alone 2.5 (2.7, 3.1) ms, and the complete filter with the altitude and swing rows 3.5 (3.8, 4.1) ms: below the 5 ms period of the full-order trials, above the 1 ms hold of Table I, which is a simulation setting. A backup filter that integrates $\pi^\sharp$ numerically [6] on the same

machine (Dormand–Prince, relative tolerance 10^{-8} , central differences with step 10^{-4} , 19 integrations per evaluation over the horizon of the maneuver plus 0.5 s) reproduced D within a relative error of $1.1 \cdot 10^{-6}$ and the commands of the filter within 0.4 % of $f_{\max}$ at 60 states, at 129 s per evaluation (median; maximum 216 s). Solver failures, model mismatch, and inter-sample excursions preclude a real-time guarantee.

VII. CONCLUSION

Cable geometry makes braking authority a controlled state that changes at a bounded rate. An initially feasible HOCBF filter can lose feasibility when the geometry cannot change quickly enough, and an explicit swing-and-brake maneuver gives continuous-time inner and outer viability bounds for every system with second-order authority. The inner bound is conservative where the cables must first swing: safe trajectories were found from as little as 0.25 D there, against 0.97–0.98 D with the cables already behind the payload; the reserved swing acceleration accounts for part of this gap.

On the exact model, the certificate holds up to the hold: with a period of 1 ms and bounded swing accelerations, one of 100 trials started at the boundary of the certified set reached the wall by $2 \mu\text{m}$, none did with 0.5 ms, and four did with 5 ms. A full-order model with slack cables and a 5 ms hold lies outside the hypotheses of the theorem, and the sampled filter then needs an empirical margin. A certified sampled-data bound, full-order trials with the altitude barriers, a less conservative swing reserve, and a distributed implementation are the next requirements for a deployment claim.

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